Agentic AI for Process Capability & Behavior Assessment

Sandbox Pilot Proposal – North River Diagnostics

Agenda

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Pain Points

- **Scrap**, **rework**, and **downtime** driven by variability in critical characteristics
- Capability and behavior assessments are **manual**, **fragmented**, and **time-consuming**
- Analysis often takes days, delaying corrective action
- Different engineers interpret the same data differently
- Leadership receives **conflicting conclusions** and **reduced confidence** in Cp/Cpk metrics

Core issue: NRD lacks a fast, standardized way to turn process data into reliable decisions.

Why the current state creates Risk?

- Stable but incapable processes require ***redesign***, not firefighting
- Unstable processes make capability indices ***unreliable***
- Today, these distinctions are not enforced ***consistently***
- Improvement resources may be applied in the ***wrong*** direction
- Scaling analysis through additional headcount is ***slow*** and ***costly***

Pilot
Objective and Focus

- Improve **speed**, **consistency**, and **scalability** of capability & stability decisions

- **Augment** engineers, **standardize** interpretations, **reduce** misdirected actions

Pilot Scope

FOCUS

- One production line or process family
- One critical characteristic
- Two representative scenarios:
  - Stable but incapable process
  - Unstable and incapable process

DECISIONS SUPPORTED

- Is the process stable enough to trust Cpk?
- Is this a capability problem or a stability problem?
- What action should be taken first?

High Level Agentic AI Workflow

Inputs: Azure Blob Storage

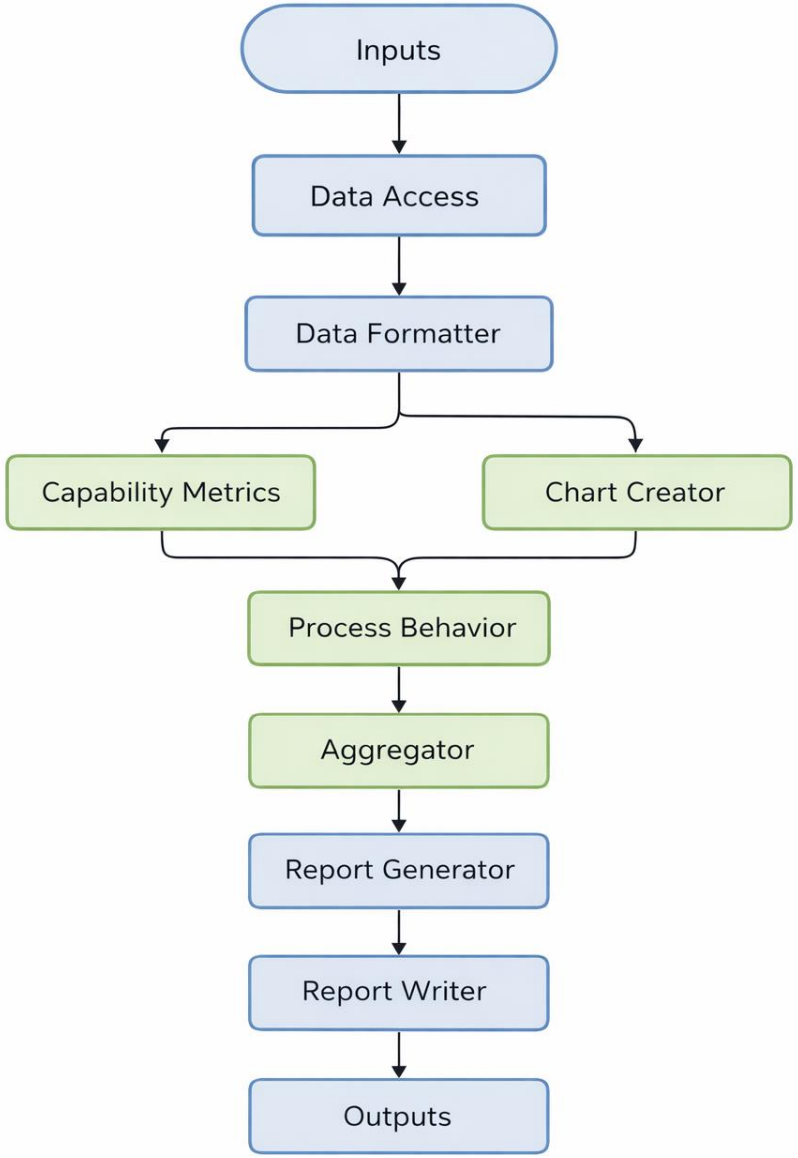
Prompt Flow Hub Agents:
- Data Formatter (LLM for normalizing process data)

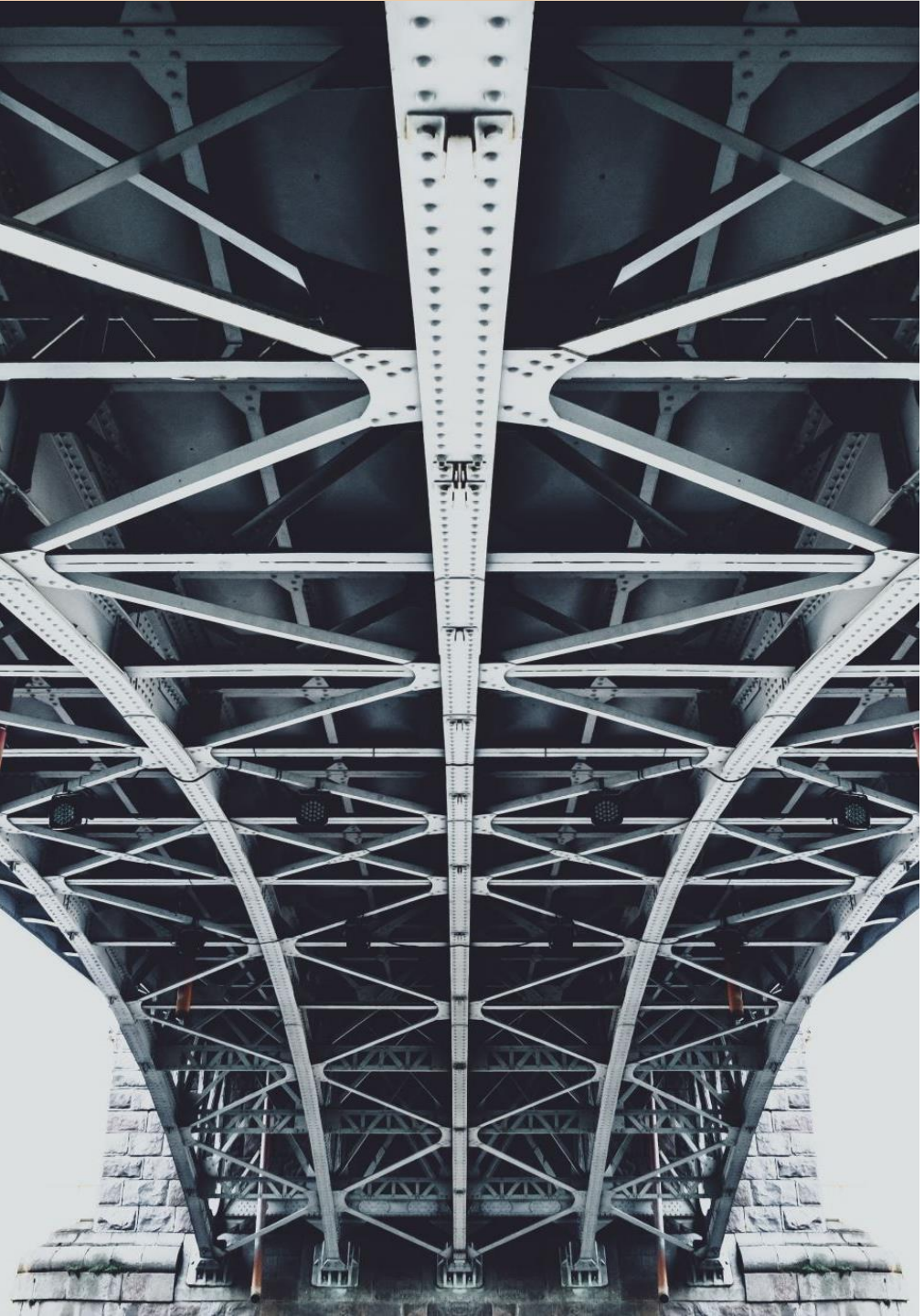
- Capability Metrics Calculator (using MCP)

- Chart Creator (using MCP)

- Process Behavior (Multimodal LLM for stability assessment)

Foundry Agents:
- Aggregator
(RAG + LLM for interpretation grounded in NRD capability guidelines)





Design Principles

- Stability assessed before capability interpretation
- Interpretations aligned to internal standards
- Aggregator Agent within Foundry Project to keep modular approach and easy modifications to RAG tools without affecting prompt flow
- Human review required for all outputs

Value Hypothesis

- **Hypothesis**

  If NRD standardizes process capability and behavior assessments using a grounded Agentic AI workflow, then engineers and leaders will make faster, more consistent, and better-directed quality decisions, without increasing headcount.

- **What changes**
  - Less time calculating and debating metrics
  - More time fixing the right problems
  - Greater confidence in reported capability

Quantified Benefits

Current Cost of the Status Quo

1. Time Cost (Manual Analysis)

Based on typical NRD workflows and dataset size (~100+ observations):

Step	Manual Effort
Data extraction & cleaning	30–45 min
Control chart setup & review	20–30 min
Capability calculations	15–30 min
Interpretation & write-up	30–45 min
Total	**~1.5–2.5 hours per process**

Even **worse, unstable** processes often require re-analysis once stability issues are discovered late.

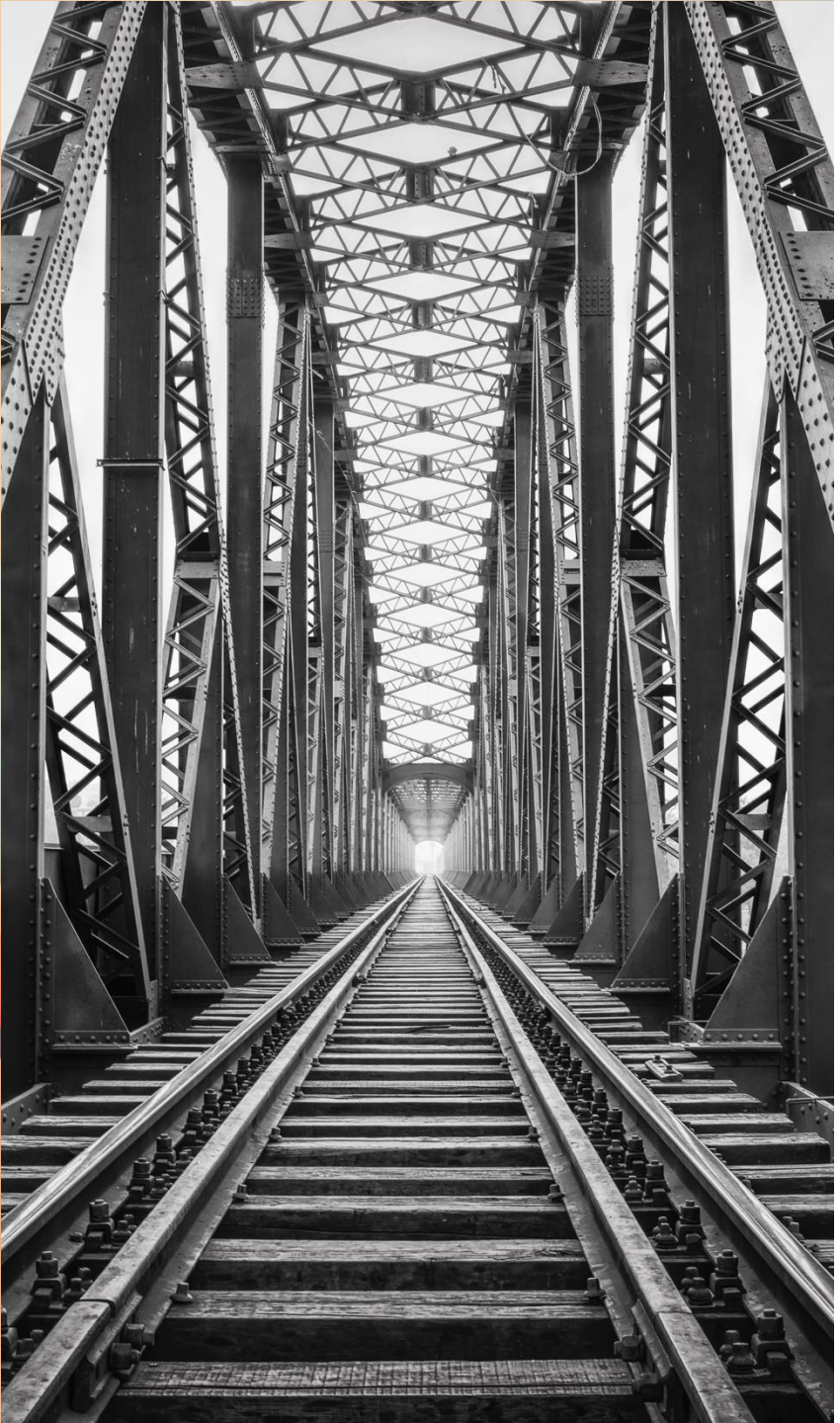
Quantified Benefits

Agentic AI–Enabled Workflow Impact

Dimension	Current State	With Agentic AI
Stability verification	Inconsistent, sometimes skipped	Enforced first step
Capability calculation	Manual, repeated	Automated via MCP
Interpretation logic	Engineer-dependent	Standardized via NRD guide
Time per analysis	1.5–2.5 hrs	~10–20 min (review-focused)
Error risk	High (misapplied Cpk)	Reduced by design
Estimated time reduction: 70–85% per analysis		

Risks & Mitigation

- **Technical risk**
  - Risk: Incorrect or misleading interpretations
  - Mitigation: MCP tools + grounding in NRD guidelines + stability gating
- **Adoption risk**
  - Risk: Lack of trust or misuse
  - Mitigation: Transparent reasoning, human-in-the-loop, standardized language
- **Governance & data security**
  - Risk: Data exposure or uncontrolled use
  - Mitigation: Sandboxed environment, limited scope, logging, no external sharing



Roadmap

- **Phase 1 (Pilot)**
  - Single line, single characteristic
  - Manual trigger, human review required
  - Success measured by speed and clarity
- **Phase 2**
  - Expand to multiple lines
  - Broader guideline coverage
  - Recurring review cadence
- **Phase 3**
  - Multi-site deployment
  - Integration with SPC and quality dashboards

Ask

- **Approve sandbox pilot**
- **Provide limited data access**
- **Endorse human-in-the-loop governance model**

This pilot is not about replacing engineering judgment. It is about **making the right decision faster, more consistently, and with less wasted effort**.